Syllabus Details

Syllabus ID:	12038
Syllabus Name:	**Object-Oriented Programming_Lập trình hướng đối tượng**
Course Name English:	**Object-Oriented Programming**
Subject Code:	**PRO192**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 85' practical exam + 104h self-study
Pre-Requisite:	Pass PRF192
Description:	-This subject introduces the student to object-oriented programming. The student learns to build reusable objects, encapsulate data and logic within a class, inherit one class from another and implement polymorphism. - Compose technical documentation of a Java program using internal comments - Adhere to object-oriented programming principles including encapsulation, polymorphism and inheritance when writing program code - Trace the execution of Java program logic to determine what a program does or to validate the correctness of a program
StudentTasks:	- Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises given by instructor in class or at home and submit on time. - Use laptop in class only for learning purpose - Promptly access to the FU FLM at https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	Text Book 'NetBean x.x Internet
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1286/QĐ-ĐHFPT dated 11/22/2024
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	11/22/2024

4 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	No
1	Java 8 Specification https://docs.oracle.com/javase/specs/jvms/se8/html/jvms-1.html#jvms-1.1						False	False	True	
2	Core Java, Volum 1: Fundamentals	Cay Horstmann	Pearson	2018	11th	978-0135166307	False	True	False	
3	Core Java, Volum 2: Advanced features	Cay Horstmann	Pearson	2019	11th	978-0135166314	False	True	False	
4	Object-Oriented Programming using Java (website (offline): Học liệu do FU biên soạn)						True	False	False	

9 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand the concepts of object oriented (OO) programs to solve problems and fundamentals of object-oriented programming in Java
2	CLO2	Practice basic Java language syntax and semantics to write Java programs and use concepts such as variables, conditional and iterative execution methods
3	CLO3	Uses streams to read and write data from/to different types of sources/targets
4	CLO4	Discuss the benefits and the use of JAVA's Exceptional handling mechanism
5	CLO5	Identify classes, objects, members of a class and relationships among them needed for a specific problem
6	CLO6	Explain the concept and demonstrates the use of Polymorphism, Encapsulation, Abstraction and Inheritance in java
7	CLO7	Discuss the principles and the use of abstract classes and interfaces in java
8	CLO8	Understand and implement a complete program using object array
9	CLO9	Explain the principles and the use of some (java collections) abstract data types (list, set, map)

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Chapter 1: Introduction 1.1 Welcome to Object-Oriented	Offline	CLO1	I	Textbook, slides	link1-1	Textbook, slides	
2	1.2 Object Terminology and Installation JDK - IDE	Offline	CLO1		Textbook, slides	link2-1	Textbook, slides	
3	Chapter 2: Basic Java language 2.1 Basic Java language: Variable, data type	Offline	CLO2	IT	Textbook, slides		Textbook, slides	
4	2.1 Basic Java language: Control statements	Offline	CLO2	IT	Textbook, slides		Textbook, slides	
5	2.1 Basic Java language: Method - function- Parameter passing	Offline	CLO2	IT	Textbook, slides		Textbook, slides	
6	2.1 Basic Java language: Array, ArrayList	Offline	CLO3	IT	Textbook, slides		Textbook, slides	
7	2.1 Basic Java language: String, StringBuffer	Offline	CLO4	IT	Textbook, slides		Textbook, slides	
8	2.2 Input and Output (Standard I/O)	Offline	CLO3	IT	Textbook, slides		Textbook, slides	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
9	2.2 Input and Output (Standard I/O) - cont'd	Offline	CLO4	IT	Textbook, slides		Textbook, slides	
10	Lab 1 assistance	Offline	CLO2,CLO3	U	lab's questions		lab's questions	
11	Lab 1 assistance	Offline	CLO2,CLO4	U	lab's questions		lab's questions	
12	2.4 Dynamic Memory	Offline	CLO4	I,T	Textbook, slides		Textbook, slides	
13	Chapter 3: OOP 3.1 OOP concept	Offline	CLO5	I,T	Textbook, slides		Textbook, slides	
14	3.2 Data abstraction - concept and technic	Offline	CLO5	I,T	Textbook, slides		Textbook, slides	
15	3.3 Class definition: data and function members	Offline	CLO5	I,T	Textbook, slides		Textbook, slides	
16	3.4 Encapsulation	Offline	CLO5, CLO6	I,T	Textbook, slides		Textbook, slides	
17	Lab 2 assistance	Offline	CLO5, CLO7	U	Lab's questions		Lab's questions	
18	Lab 2 assistance	Offline	CLO5, CLO8	U	Lab's questions		Lab's questions	
19	3.5 Inheritance: Inheritance introduction	Offline	CLO5,CLO6	I, T	Textbook, slides		Textbook, slides	
20	3.6 Inheritance: Derived Classes	Offline	CLO5,CLO7	I, T	Textbook, slides		Textbook, slides	
21	Lab 3 assistance	Offline	CLO5,CLO8	U	Textbook, slides		Textbook, slides	
22	Lab 3 assistance	Offline	CLO5,CLO9	U	Textbook, slides		Textbook, slides	
23	3.7 Polymorphism: Polymorphism concepts	Offline	CLO7	I, T	Textbook, slides		Textbook, slides	
24	3.8 Override and overriden methods	Offline	CLO7	I, T	Textbook, slides		Textbook, slides	
25	Lab 4 assistance	Offline	CLO7	U	Textbook, slides		Textbook, slides	
26	Lab 4 assistance	Offline	CLO7	U	Textbook, slides		Textbook, slides	
27	Chapter 4: Abstract class - Interface 4.1 Abstract class	Offline	CLO7	IT	Textbook, slides		Textbook, slides	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
28	4.1 Abstract class - con't	Offline	CLO7	IT	Textbook, slides		Textbook, slides	
29	4.2 Interface	Offline	CLO7	IT	Textbook, slides		Textbook, slides	
30	4.2 Interface - cont'd	Offline	CLO7	IT	Textbook, slides		Textbook, slides	
31	Chapter 5: 5.1 Error & Exception Concepts	Offline	CLO4		Textbook, slides		Textbook, slides	
32	5.2 Handling Exception	Offline	CLO4		Textbook, slides		Textbook, slides	
33	Progress test 1+ workshop evaluation	Offline	CLO1,CLO2,CLO3,CLO4,CLO5,CLO6,CLO7		Textbook, slides,assignment's questions		Textbook, slides,assignment's questions	
34	Progress test 1+ workshop evaluation	Offline	CLO1,CLO2,CLO3,CLO4,CLO5,CLO6,CLO7	I,T	Textbook, slides,assignment's questions		Textbook, slides,assignment's questions	
35	Chapter 6: Array of Objects 6. Basic Operators: add, update, remove, sort, find	Offline	CLO8	IT	Textbook, slides, lab's questions		Textbook, slides, lab's questions	
36	Array of Objects: 6. Basic Operators: add, update, remove, sort, find - cont'd	Offline	CLO9	IT	Textbook, slides, lab's questions		Textbook, slides, lab's questions	
37	Array of Objects: 6. Basic Operators: add, update, remove, sort, find - cont'd	Offline	CLO9	IT	Textbook, slides, lab's questions		Textbook, slides, lab's questions	
38	Array of Objects: 6. Basic Operators: add, update, remove, sort, find - cont'd	Offline	CLO9	IT	Textbook, slides, lab's questions		Textbook, slides, lab's questions	
39	Chapter 7: Collections in Java: 7.1 Overview	Offline	CLO9	I,T	Textbook, slides		Textbook, slides	
40	7.2 List	Offline	CLO9	I,T	Textbook, slides		Textbook, slides	
41	7.3 Set	Offline	CLO9	I,T	Textbook, slides		Textbook, slides	
42	7.3 Set	Offline	CLO9	I,T	Textbook, slides		Textbook, slides	
43	7.4 Map	Offline	CLO9	I,T	Textbook, slides		Textbook, slides	
44	7.4 Map	Offline	CLO9	I,T	Textbook, slides		Textbook, slides	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
45	Lab 5 assistance	Offline	CLO4,CLO5,CLO6,CLO7	U	Lab's questions		Lab's questions	
46	Lab 5 assistance	Offline	CLO4,CLO5,CLO6,CLO7	U	Lab's questions		Lab's questions	
47	Chapter 8: Input and Output (File I/O)	Offline	CLO4, CLO5, CLO6, CLO7, CLO8	IT	Textbook, slides, assignment's questions		Textbook, slides, assignment's questions	
48	8. Input and Output (File I/O) - cont't	Offline	CLO4, CLO5, CLO6, CLO7, CLO8	IT	Textbook, slides, assignment's questions		Textbook, slides, assignment's questions	
49	8. Input and Output (File I/O) - cont't	Offline	CLO4, CLO5, CLO6, CLO7, CLO8	IT	Textbook, slides, assignment's questions		Textbook, slides, assignment's questions	
50	8. Input and Output (File I/O) - cont't	Offline	CLO4, CLO5, CLO6, CLO7, CLO8	IT	Textbook, slides, assignment's questions		Textbook, slides, assignment's questions	
51	Lab 6 assistance	Offline	CLO9	T,U	lab's questions		lab's questions	
52	Lab 6 assistance	Offline	CLO9	T,U	lab's questions		lab's questions	
53	Progress test 2+ workshop evaluation	Offline	CLO8, CLO9	U	Textbook, slides		Textbook, slides	
54	Progress test 2+ workshop evaluation	Offline	CLO8, CLO9	U	Textbook, slides		Textbook, slides	
55	Assignment Evaluation	Offline		I,T	Textbook, slides		Textbook, slides	
56	Assignment Evaluation	Offline		I,T	Textbook, slides		Textbook, slides	
57	Review	Offline	All LOs					
58	Review	Offline	All LOs					
59	Review	Offline	All LOs					
60	Review	Offline	All LOs					

52 Constructive question(s)

No.	Session No	Name	Details
1	1	CQ1.1	Programming Language and Programming Paradigm Hints: What is a programming language? Lists at least 2 programming languages? What is a programming paradigm? Lists at least 2 programming paradigms?
2	1	CQ1.2	Structured Programming Paradigm Hints: What are characteristics of Structured Programming Paradigm (review Programming Fundamentals (PRF))? What problems does the PRF solve? What problems is difficult for the PRF to solve?
3	2	CQ1.3	Object-oriented Programming (OOP) Paradigm Hints: What are the pros and cons of OOP over PRF? Look up the code available to illustrate to highlight this difference.
4	2	CQ1.4	What are the concepts of major features in OOP (Abstraction, Encapsulation, Inheritance, Polymorphism)? Hints: Concept in the real world. Example. Concept in the programming
5	3	CQ1.5	How may Data types can be use? Why we need variables? Which operators can we use in Java? Hints for Data type, variable, operators (logic, bitwise ...) List of data types Variable naming Convert data type (auto, command) Syntax and example Scope of variable Evaluating Expressions and Operator Precedence

No.	Session No	Name	Details
6	4	CQ2.1	What are basic statements in Java? What are their usecase and syntax? Hints: Selection (If else, Switch case) Loop (for, while, do..while)
7	5	CQ2.2	What are Array and String types? When we use them? What are their syntax? Hints for Array and String type Syntax and example
8	6	CQ2.3	What are the basic input output? Give examples Hints for Input/output data (keyboard, screen): Library Syntax and example
9	7	CQ3.1	What are streams? Describe about Input output stream? Hints: Concept Input/Output stream
10	8	CQ3.2	How to access an manage directories and files, text files? Hints: Syntax and example to access
11	9	CQ3.3	Whar are Binary files? Hints: Syntax and example to access
12	10	CQ4.1	How to distinguish Text, UTF, and Unicode files? Hints: Concept, Use Case Syntax and example
13	12	CQ4.2	What is exception? Hints: What is an exception? Why do we use exception? Where do we use exception?
14	13	CQ5.1	How to use try catch – finally blocks? Hints: Syntax and example
15	14	CQ5.2	What are throw, throws? Hints: Concept, Use Case Syntax and example
16	15	CQ5.3	When do we need assertion? Hints for Assertions: Concept, Use Case Syntax and example
17	17	CQ6.1	What are Class and Object? How to distinguish them? Hints for Class and Object: Class, Object in the real world Class, Object in programming Class vs. Object.
18	18	CQ6.2	What are constructor methods? Hints: Concept. Use case Syntax and code example Implement case? How many constructors should be implemented? What should we will write in constructor's body?
19	19	CQ7.1	What "this" keyword means? Hints: Concept Use case Syntax and code example
20	20	CQ7.2	What is UML? Hints: Defines the symbols used in UML Explain an existing class diagram Build a class diagram
21	21	CQ7.3	What are member methods? Hints: Concept. Use case, Syntax and code example, Implement case? What should we will write in member method's body
22	22	CQ8.1	How to distinguish member functions vs. Constructors? Hints: Concept, Use Case Syntax and example
23	23	CQ8.2	What are Parameter and Argument? Hints: Concept, Example
24	24	CQ8.3	How many access levels in OOP? What are they? Hints: List access levels? Explain why and why not for class and each class member form? Syntax and code example for each situation
25	26	CQ9.1	What is Inheritance in OOP? Hints: Use case Syntax and example
26	27	CQ9.2	What should we concern when casting? Hints: How instance object variable casting? instance object variables vs. primitive data type variables
27	28	CQ10.1	What is "instanceof" operator? Hints: Use case Syntax and example
28	29	CQ10.2	How to distinguish superclass and Subclass? Hints: Concept, Use Case Syntax and example
29	30	CQ10.3	What are access levels between members of Superclass, Subclass, and from the instance object? Hints: Concept, Use Case Syntax and example
30	31	CQ11.1	How the parent constructor called? (from instance object, from subclass) Hints: Concept, Use Case Syntax and example
31	32	CQ11.2	What are differences between "this" and "super" Hints: Concept, Use Case Syntax and example
32	34	CQ12.1	What is abstract class? Hints: Concept Syntax and example Abstract class vs. normal class
33	35	CQ12.2	What are members of abstract class? Hints: List member types in an abstract class Use case Syntax and code example Which can / cannot be member of abstract class? Why?
34	36	CQ12.3	What are access levels in abstract class? Hints: Are public, protected, private and default can / cannot set to the abstract class and its members? Syntax and code example.
35	37	CQ13.1	What are anonymous class and nest class? Hints: Anonymous Class (Concept, Use case, Syntax and example) Nest Class (Concept, Use case, Syntax and example)
36	38	CQ13.2	What is interface? Hints: Concept Syntax and example
37	39	CQ13.3	What are Interface members? Hints: List member types in an interface Default method Use case Syntax and code example Which can / cannot be member of interface? Why?
38	40	CQ14.1	What are differences among Interface, Abstract Class and Class? Hints: Use case Syntax and code example

No.	Session No	Name	Details
39	41	CQ14.2	Terms and definitions in java OOP? Hints: Static, abstract, finally, public, private, protected .... Use case and example
40	42	CQ14.3	What is Polymorphism? Hints: Concept, Use Case
41	43	CQ15.1	What is Overloading? Hints: Concept, Use Case Syntax and example
42	44	CQ15.2	What is Overriding? Hints: Concept, Use Case Syntax and example
43	47	CQ16.1	What are Array of Objects? Hints: Concept, Use Case Syntax and example
44	48	CQ16.2	What are differences between Array of Objects vs. Array Variables? Hints: Concept, Use Case Syntax and example
45	49	CQ17.1	What are differences between Array Arguments vs. Array Parameters? Hints: Concept, Use Case Syntax and example
46	51	CQ17.2	How to input an object into a file? Hints for Object input File Concept, Use Case Syntax and example
47	52	CQ18.1	How to get an object from a file? Hints: Concept, Use Case Syntax and example
48	56	CQ19.1	What is Collection? Hints: Concept, Structure Use case Syntax and example
49	57	CQ19.2	What is List? Hints: Concept, Structure Use case Syntax and example List of method help you to control a list
50	58	CQ20.1	What is Set? Hints: Concept, Structure Use case Syntax and example List of method help you to control a set
51	59	CQ20.2	What are differences between Set and List Hints: Concept, Use Case Syntax and example
52	60	CQ20.3	What are differences among Map, Set and List? Hints: Concept, Use Case Syntax and example

5 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	1	20.0%	>0	Option 1: 28 slots; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO1, LO2, LO3, LO4, LO5, LO6, LO7, LO8, LO9	Option 1: N/A; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: N/A; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	All subjects in syllabus	in class, by teacher	
2	Lab	on-going	6	10.0%	>0	Option 1: 90'/each; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO1, LO2, LO3, LO4, LO5, LO6, LO7, LO8, LO9	Option 1: N/A; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: N/A; Option 2 (For Constructivism Approach only): Follow lecturer's proposal		in class, by teacher	
3	Practical Exam	on-going	1	30.0%	>0	85'	LO1->LO9				By exam board	

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
4	Progress Test	on-going	2	10.0%	>0	Option 1: 30'/each; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO1, LO2, LO3, LO4, LO5, LO6	Option 1: Multiple choices Marked by Computer or a suitable format; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 20/each; Option 2 (For Constructivism Approach only): Follow lecturer's proposal		in class, by LMS system	Instruction and schedules for Progress Tests must be presented in the Course Implementation Plan approved by director of the campus. Progress test must be taken right after the last lectures of required material. Instructor has responsibility to review the test for students after graded.
5	Final Exam	Final exam	1	30.0%	4	60'/each	LO1->LO9	Multiple choices Marked by Computer	50	All subjects in syllabus	By Exam Board	The exam questions must be updated or different at least 70% to the previous ones.